CLASS 8 · ICSE · SELINA CONCISE MATHEMATICS · SHORT NOTES

Chapter 4: Cubes and Cube Roots

Topic-wise short notes with solved examples

Perfect cubes and their properties, finding cube roots by prime factorisation, cube roots of negative numbers and fractions, and related problems.

1 Cubes

- The **cube** of a number is the number multiplied by itself three times: $n^3 = n \times n \times n$.
- A natural number that is the cube of a natural number is a **perfect cube**: 1, 8, 27, 64, 125, ...
- In the prime factorisation of a perfect cube, every prime occurs a number of times that is a **multiple of 3**.

n	1	2	3	4	5	6	7	8	9	10
n^3	1	8	27	64	125	216	343	512	729	1000
n	11	12	13	14	15	16	17	18	19	20
n^3	1331	1728	2197	2744	3375	4096	4913	5832	6859	8000

2 Properties of cubes

- The cube of an even number is even and the cube of an odd number is odd.
- The cube of a **negative** number is negative: $(-4)^3 = -64$.
- The cube of a fraction: $\left(\frac{a}{b}\right)^3 = \frac{a^3}{b^3}$.
- The units digit of n^3 depends only on the units digit of n:

Units digit of n	0	1	2	3	4	5	6	7	8	9
Units digit of n^3	0	1	8	7	4	5	6	3	2	9

- A number ending in k zeros has a cube ending in $3k$ zeros.
- **Odd-number pattern**: $1 = 1^3$, $3 + 5 = 2^3$, $7 + 9 + 11 = 3^3$, $13 + 15 + 17 + 19 = 4^3$.
- $1^3 + 2^3 + \ldots + n^3 = (1 + 2 + \ldots + n)^2$.

Example 2.1 Find the cube of -7, of $\frac{2}{5}$ and of 0.3.

SOLUTION

-343, $\frac{8}{125}$ and 0.027

Example 2.2 What is the units digit of the cube of 2387?

SOLUTION

The units digit 7 gives a cube ending in 3.

3

Example 2.3 Find $1^3 + 2^3 + 3^3 + 4^3 + 5^3$ using a pattern.

SOLUTION

$(1 + 2 + 3 + 4 + 5)^2 = 15^2$

225

3 Testing for a perfect cube

- Write the prime factorisation and group equal primes in **triplets**. If no prime is left over, the number is a perfect cube.
- To make a number a perfect cube, **multiply** by the primes needed to complete the triplets, or **divide** by the primes left over.

Example 3.1 Is 2744 a perfect cube?

SOLUTION

$2744 = 2^3 \times 7^3$

Yes; it is 14^3

Example 3.2 Find the smallest number by which 1323 must be multiplied to get a perfect cube.

SOLUTION

$1323 = 3^3 \times 7^2$. One more 7 is needed.

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Example 3.3 Find the smallest number by which 8640 must be divided to get a perfect cube.

SOLUTION

$8640 = 2^6 \times 3^3 \times 5$

5; the quotient 1728 is 12^3

4 Cube roots

- If $x^3 = y$, then x is the **cube root** of y: $x = \sqrt[3]{y}$.
- **Prime factorisation method**: group the prime factors in triplets, take one from each triplet and multiply.

- $\sqrt[3]{-a} = -\sqrt[3]{a}$. The cube root of a negative number is negative.
- $\sqrt[3]{ab} = \sqrt[3]{a} \times \sqrt[3]{b}$ and $\sqrt[3]{\frac{a}{b}} = \frac{\sqrt[3]{a}}{\sqrt[3]{b}}$.

Example 4.1 Find $\sqrt[3]{9261}$.

SOLUTION

$9261 = 3^3 \times 7^3$

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Example 4.2 Find $\sqrt[3]{-5832}$.

SOLUTION

$5832 = 2^3 \times 3^6$, so $\sqrt[3]{5832} = 2 \times 9 = 18$.

-18

Example 4.3 Find $\sqrt[3]{\frac{-125}{343}}$ and $\sqrt[3]{3\frac{3}{8}}$.

SOLUTION

$\frac{-5}{7}$; and $3\frac{3}{8} = \frac{27}{8}$

$\frac{-5}{7}$ and $\frac{3}{2}$

Example 4.4 Find $\sqrt[3]{0.008}$ and $\sqrt[3]{15.625}$.

SOLUTION

$0.008 = \frac{8}{1000}$ and $15.625 = \frac{15625}{1000}$, with $25^3 = 15625$

0.2 and 2.5

Example 4.5 Find $\sqrt[3]{27 \times 64}$.

SOLUTION

3×4

12

5 Cube root by inspection

- For a perfect cube of up to six digits: the **units digit** of the root comes from the units digit of the number (using the table). The **tens digit** comes from the digits left after removing the last three: find the largest cube not greater than them.

Example 5.1 Find the cube root of 17576 by inspection.

SOLUTION

It ends in 6, so the root ends in 6. Removing the last three digits leaves 17: $2^3 = 8 \leq 17 < 27$.

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Example 5.2 Find the cube root of 110592 by inspection.

SOLUTION

Ends in 2, so the root ends in 8. The remaining digits 110: $4^3 = 64 \leq 110 < 125$.

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6 Word problems

- Volume of a cube of edge a is a^3, so edge $= \sqrt[3]{\text{volume}}$.

Example 6.1 The volume of a cubical box is 13824 cu cm. Find its edge and the area of one face.

SOLUTION

$13824 = 2^9 \times 3^3$, so the edge is $2^3 \times 3 = 24$ cm.

24 cm; 576 sq cm

Example 6.2 Three numbers are in the ratio $1 : 2 : 3$ and the sum of their cubes is 972. Find them.

SOLUTION

$x^3 + 8x^3 + 27x^3 = 36x^3 = 972$, so $x^3 = 27$.

3, 6 and 9

7 Common mistakes

- Writing $n^3 = 3n$.
- Making pairs instead of triplets when finding a cube root.
- Saying the cube root of a negative number does not exist. It is negative.
- Writing $\sqrt[3]{a+b} = \sqrt[3]{a} + \sqrt[3]{b}$.
- Wrong decimal places: $\sqrt[3]{0.008} = 0.2$, not 0.02.
- Using the rule "squares never end in 2, 3, 7, 8" for cubes. A cube can end in any digit.